

Homework 2

1.3. See Figure P1.3, the input voltage is $u_i = 5 \sin(\omega t)$ the diode's working voltage is $U_D = 0.7 \text{ V}$, ask: please draw the voltage transmission curve: output voltage as function of input voltage ($u_o \sim u_i$).

1.3 电路如图 P1.3 所示，已知 $u_i = 5 \sin \omega t (V)$ ，二极管导通电压 $U_D = 0.7V$ 。试画出 u_i 与 u_o 的波形图，并标出幅值。

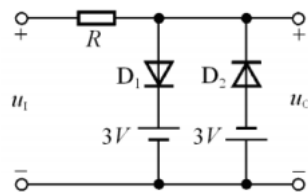


图 P1.3

1.6 As shown in Figure P1.6, voltage regulator (zener) diode's stable voltage is $U_Z = 6V$, the minimum stable current value allowed is $I_{Zmin} = 5 \text{ mA}$, and the maximum stable current value allowed is $I_{Zmax} = 25 \text{ mA}$, ask:

(1) If input voltage $U_i = 10V$, then $U_o = ?$

$$U_i = 15V, U_o = ?$$

$$U_i = 35V, U_o = ?$$

(2) If $U_i = 35 \text{ V}$ and load resistor R_L is open, what's going to happen and explain the reason?

1.6 已知图 P1.6 所示电路中稳压管的稳定电压 $U_Z = 6V$ ，最小稳定电流

$I_{Zmin} = 5mA$ ，最大稳定电流 $I_{Zmax} = 25mA$ 。

(1) 分别计算 U_i 为 $10V$ 、 $15V$ 、 $35V$ 三种情况下输出电压 U_o 的值；

(2) 若 $U_i = 35V$ 时负载开路，则会出现什么现象？为什么？

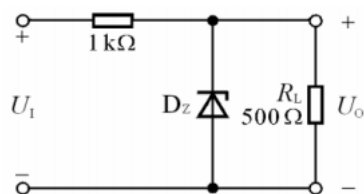


图 P1.6

1.8 We have two transistors in amplifier circuits as shown in Figure P1.8 (a) and (b). In each transistor, currents from two terminals are known. Ask:

- (1) The value and direction of the current from the third terminal;
- (2) Draw the symbols of transistors in the circle, and calculate their current amplification factors β .

1.8 现测得放大电路中两只管子两个电极的电流如图 P1.8 所示。分别求另一电极的电流，标出其方向，并在圆圈中画出管子，且分别求出它们的电流放大系数 β 。

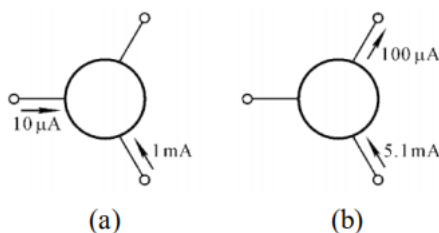


图 P1.8

- 1.9 (1) Draw the symbols of transistors in the circle;
- (2) For each transistor, it is made from silicon or the germanium?

1.9 测得放大电路中六只晶体管的直流电位如图 P1.9 所示。在圆圈中画出管子，并说明它们是硅管还是锗管。

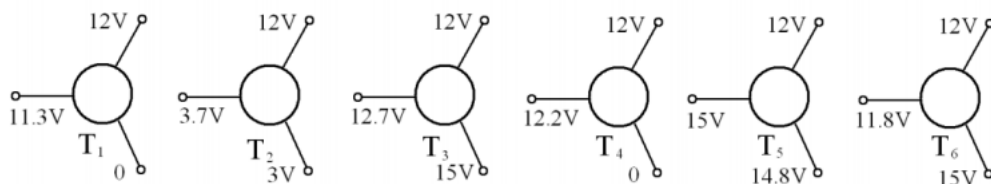


图 P1.9

1.10 See Figure P1.10, we have $U_{BE} = 0.7V$, $\beta = 50$. Ask:

- (1) When $V_{BB}=0V$, please analyze the operation modes/status of the transistor (amplification, cut-off or saturation), and the value of $u_o=?$
- (2) When $V_{BB}=1V$, please analyze the operation modes/status of the transistor (amplification, cut-off or saturation), and the value of $u_o=?$
- (3) When $V_{BB}=3V$, please analyze the operation modes/status of the transistor (amplification, cut-off or saturation), and the value of $u_o=?$

1.10 电路如图 P1.10 所示，晶体管导通时 $U_{BE} = 0.7V$ ， $\beta=50$ 。试分析 V_{BB} 为 0V、

1V、3V 三种情况下 T 的工作状态及输出电压 u_o 的值。

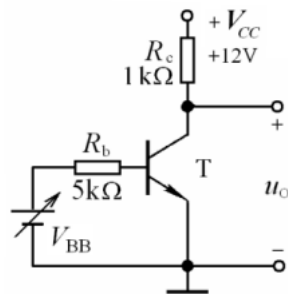


图 P1.10

- 描述 PNP 三极管工作原理（原理图），尤其是结正反偏与载流子行为的关系，三种工作状态，电学符号，电流关系（放大系数）；

Describe the working principle of PNP triode (schematic diagram), especially the relationship between positive and negative deviation and carrier behavior, three working state, electrical symbol, current relationship (amplification coefficient)